Building a RAG app to chat with your data

aka.ms/rag-azure-slides

About me

Python Cloud Advocate at Microsoft

Formerly: UC Berkeley, Khan Academy, Woebot, Coursera,Google

Find me online at:

@pamelafox

pamelafox.org

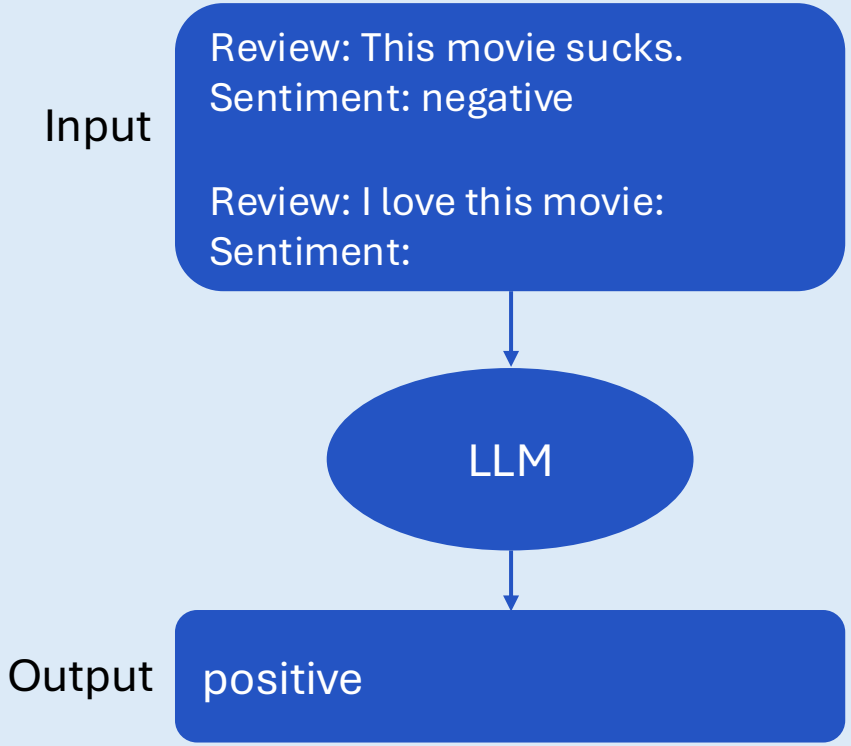
Today we'll discuss…

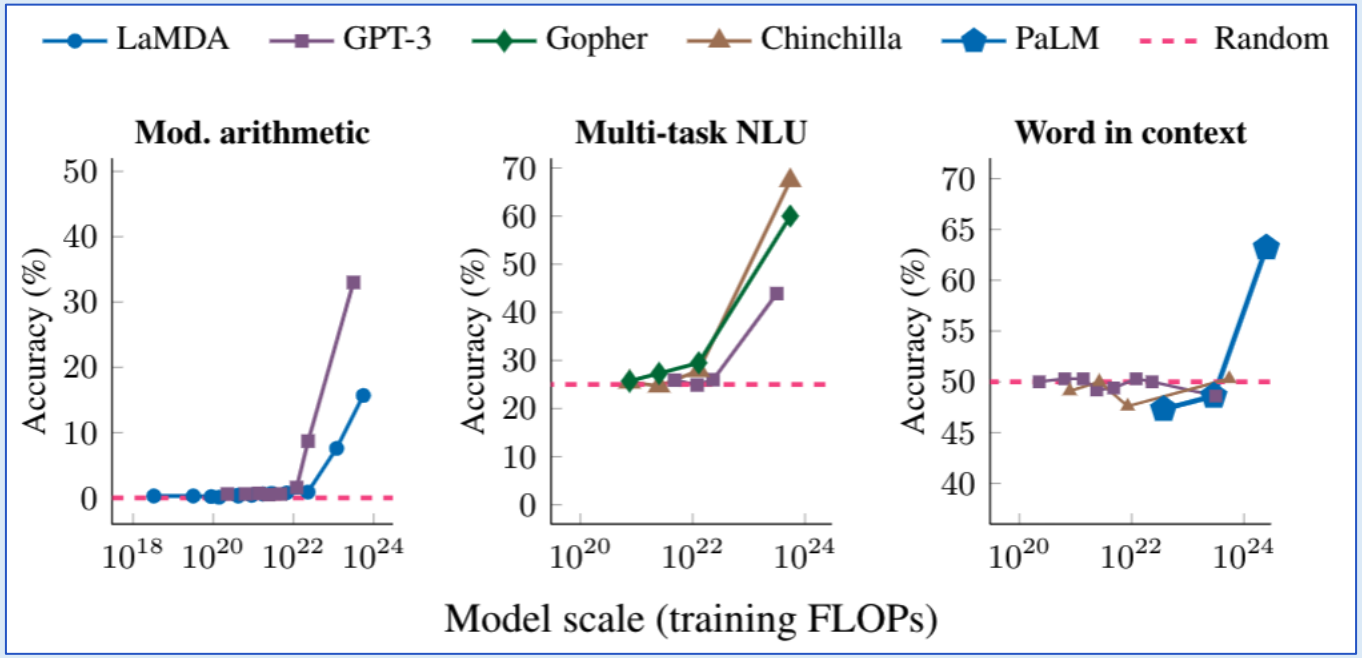
- Large Language Models
- RAG: Retrieval Augmented Generation
- Deep dive: RAG chat app solution
- Evaluating RAG apps
- Observability for RAG apps

LLMs

LLM: Large Language Model

An LLM is a model that is so large that it achieves general-purpose language understanding and generation.





LLMs in use today

Model	# of Parameters	Creator	Uses
GPT 3.5	175 B	OpenAI	ChatGPT, Copilots, APIs
GPT 4	Undisclosed	OpenAI	
PaLM	540 B	Google	Bard
Gemini	Undisclosed	Google	
Claude 2,3	130 B	Anthropic	APIs
LlaMA	70 B	Meta	OSS
Mistral-7B, Mixtral	7 B	Mistral AI	OSS

GPT: Generative Pre-trained Transformer

GPT models are LLMs based on Transformer architecture from "Attention is all you need" paper

Learn more:

- Andrej Karpathy: State of GPT
- Andrej Karpathy: Let's build GPT: from scratch, in code

Using OpenAI GPT models: Azure Studio

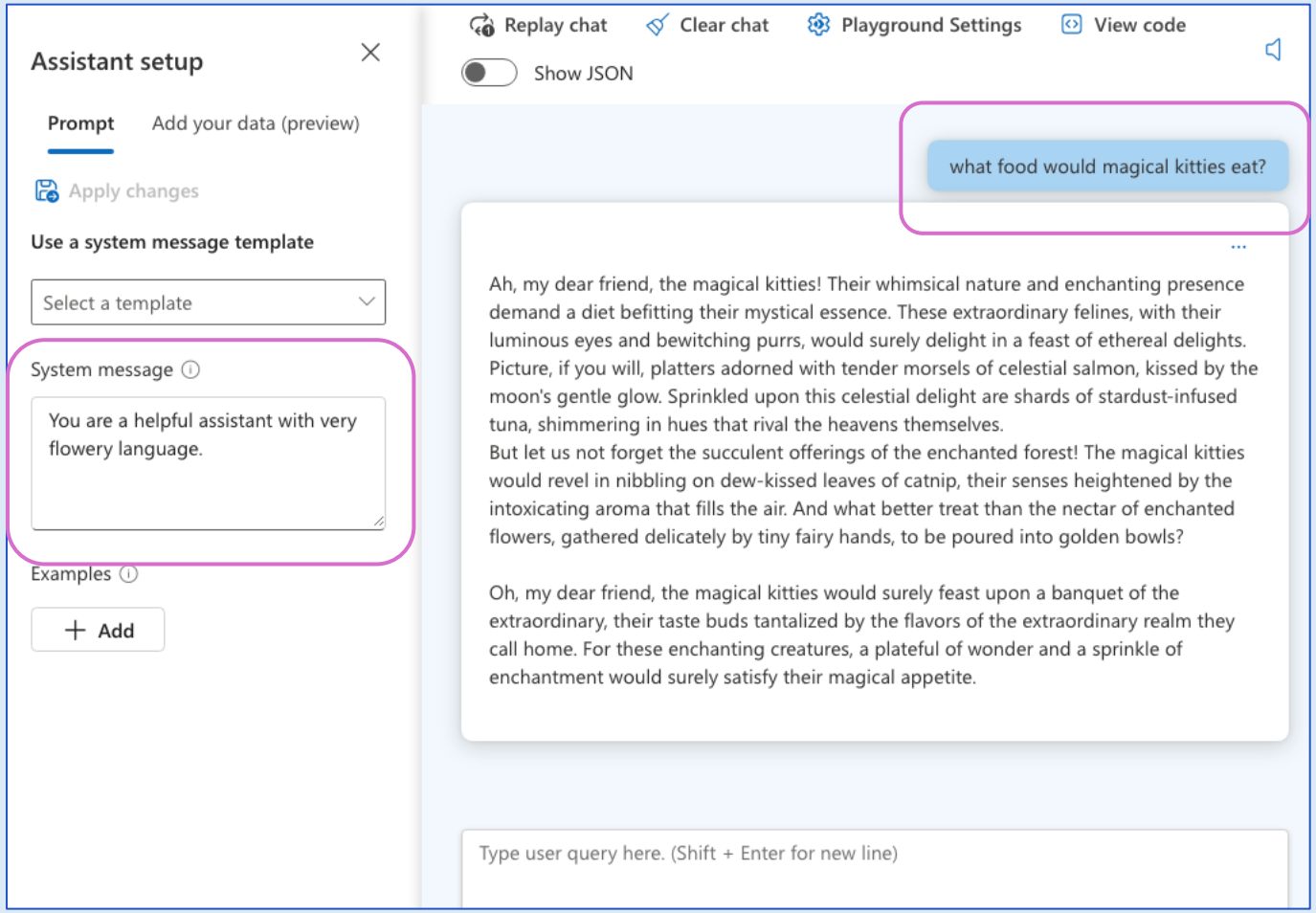
System message

+

User question

=

Chat Completion response



Using OpenAI GPT models: Python

```
response = openai.chat.completions.create(
  stream=True,
  messages = [
    {
     "role": "system",
     "content": "You are a helpful assistant with very flowery language"
    },
    {
     "role": "user",
     "content": "What food would magical kitties eat?”
    }
])

for event in response:
  print(event.choices[0].delta.content)
```

The limitations of LLMs

You

Write a model class using latest version of Flask SQL Alchemy

Assistant

Sure! Here's an example of a model class using the latest version of Flask SQLAlchemy:

```
from flask_sqlalchemy import SQLAlchemy

db = SQLAlchemy()

class User(db.Model):
    __tablename__ = 'users'

    id = db.Column(db.Integer, primary_key=True)
    username = db.Column(db.String(50), unique=True)
    email = db.Column(db.String(120), unique=True)
```

Outdated public knowledge

You

Do my company perks cover underwater activities?

Assistant

To determine whether your company perks cover underwater activities, you should consult your employee benefits package or reach out to your human resources department. They will be able to provide you with detailed information regarding the specific perks and benefits offered by your company, including any coverage for underwater activities.

No internal knowledge

Incorporating domain knowledge

Prompt engineering

In-context learning

Fine tuning

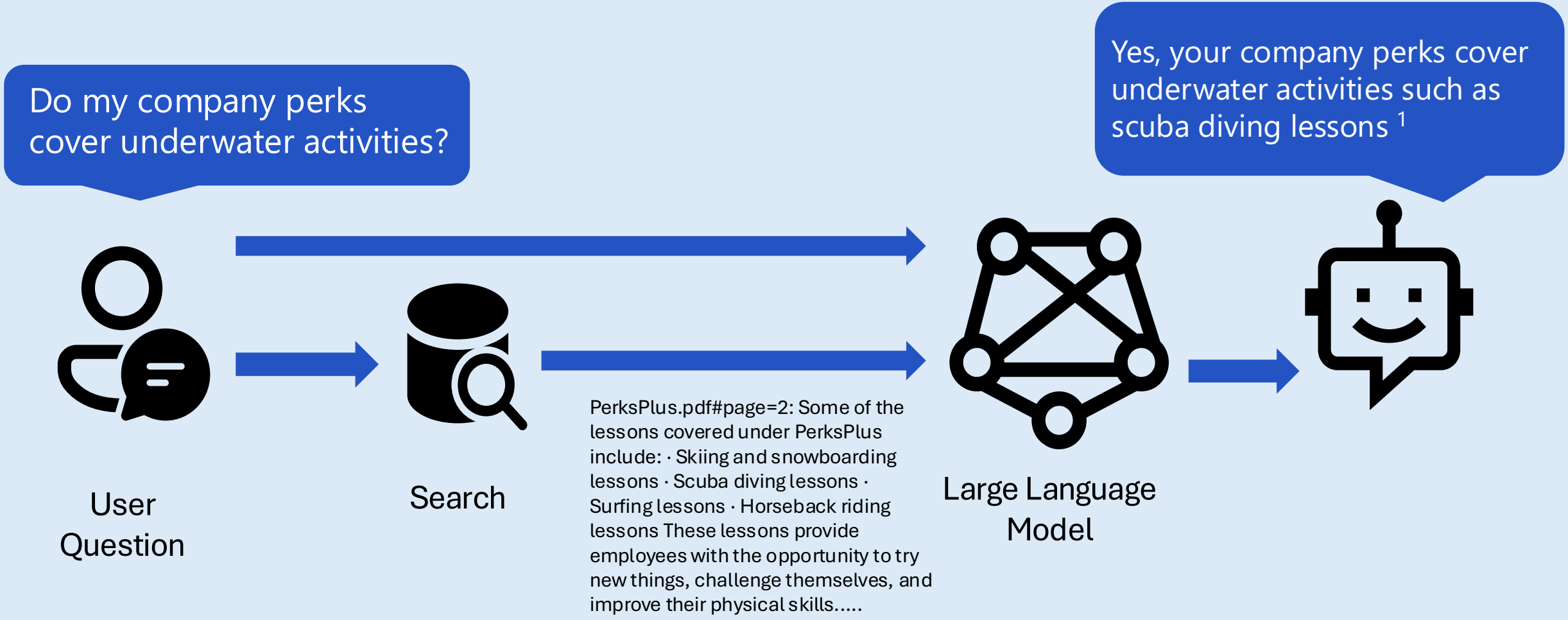
Learn new skills (permanently)

Retrieval Augmented Generation

Learn new facts (temporarily)

Retrieval Augmented Generation

RAG: Retrieval Augmented Generation



The benefit of RAG

Up-to-date public knowledge

Write a model class using lastest version of Flask SQLAlchemy

Here's a model class using the latest version of Flask SQLAlchemy:

```python
from flask import Flask
from flask_sqlalchemy import SQLAlchemy
from sqlalchemy.orm import DeclarativeBase

class Base(DeclarativeBase):
pass

db = SQLAlchemy(model_class=Base)

class YourModel(db.Model):
# Define your model fields and relationshi
pass
```

Internal (private) knowledge

Do my company perks cover underwater activities?

Yes, your company perks cover underwater activities such as scuba diving lessons [1].

Citations: 1. PerksPlus.pdf#page=2

Brand-specific knowledge

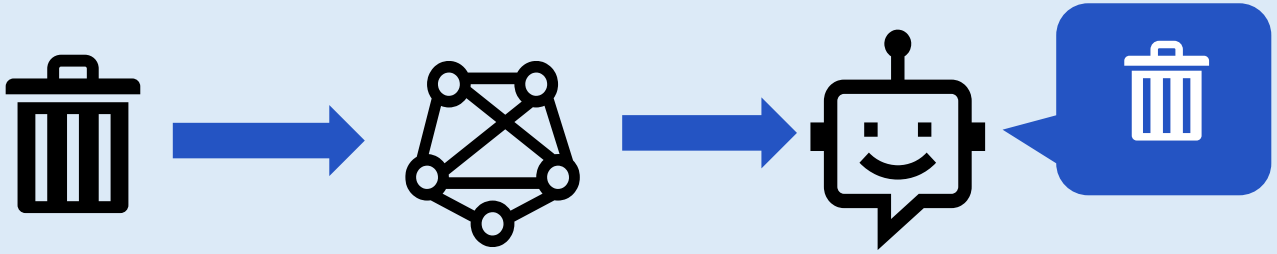
best shoes?

The best shoes for climbing would be the Vertical Journey Climbing Shoes from WildRunner [1]. For hiking, the Trail Breeze Hiking Shoes by AirStrider would be a great choice [2]. Another option for hiking would be the TrailTracker Hiking Shoes by AirStrider [3].

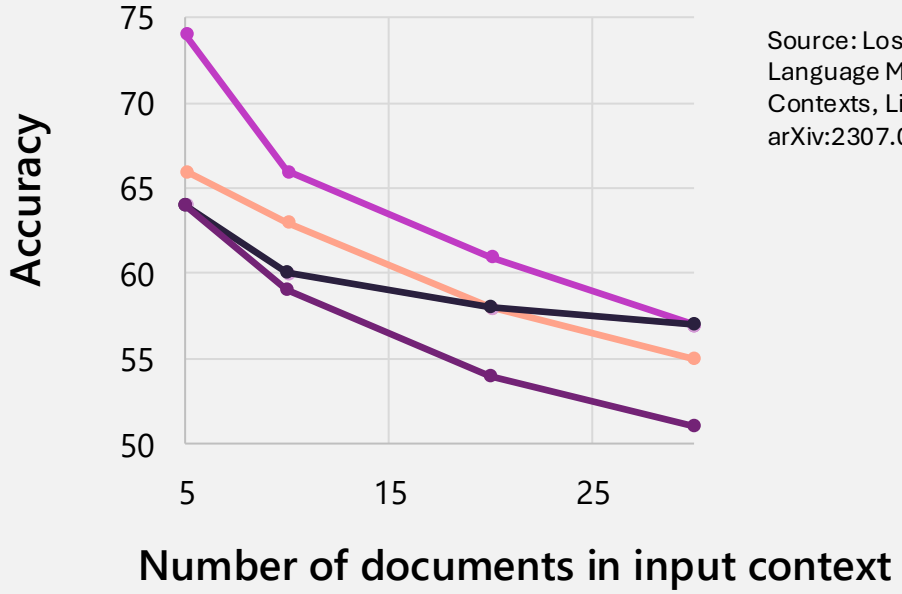
Citations: 1. Vertical Journey Climbing Shoes 2. Trail Breeze Hiking Shoes 3. TrailTracker Hiking Shoes

The importance of the search step

Garbage in, garbage out:
If the search results don't contain a good answer, the LLM will be unable to answer or will answer wrongly.



Noisy input:
If the LLM receives too much information, it may not find the correct answer amidst the noise.



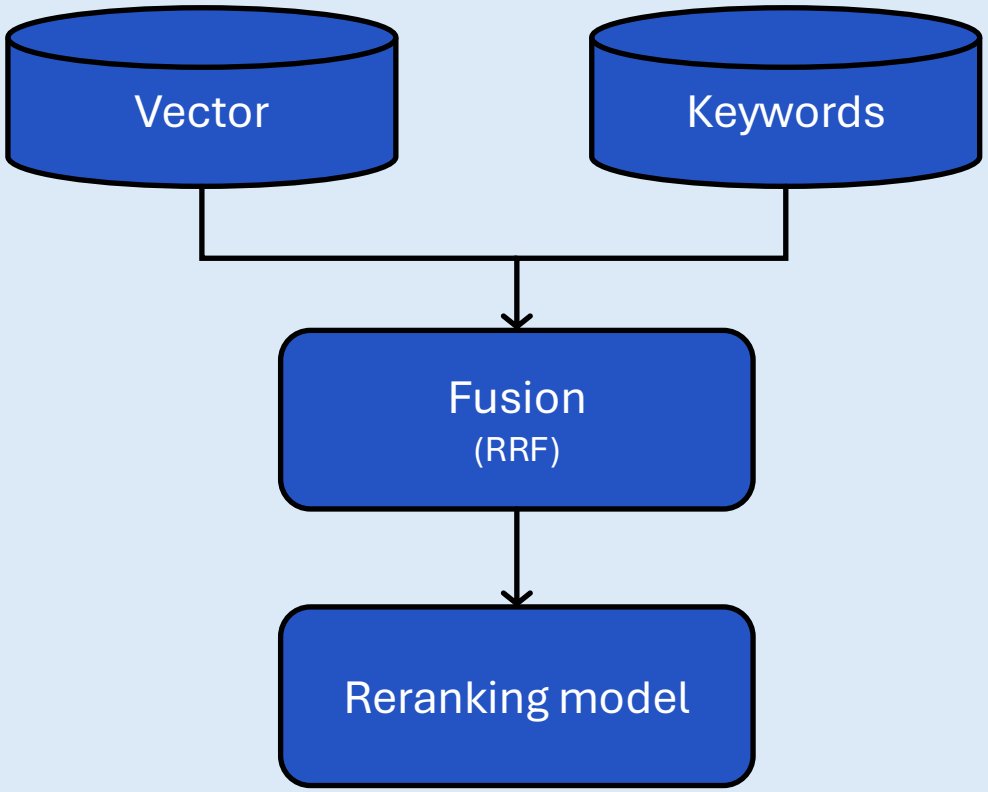
Source: Lost in the Middle: How Language Models Use Long Contexts, Liu et al. arXiv:2307.03172

Optimal search strategy

Vector search is best for finding semantically related matches

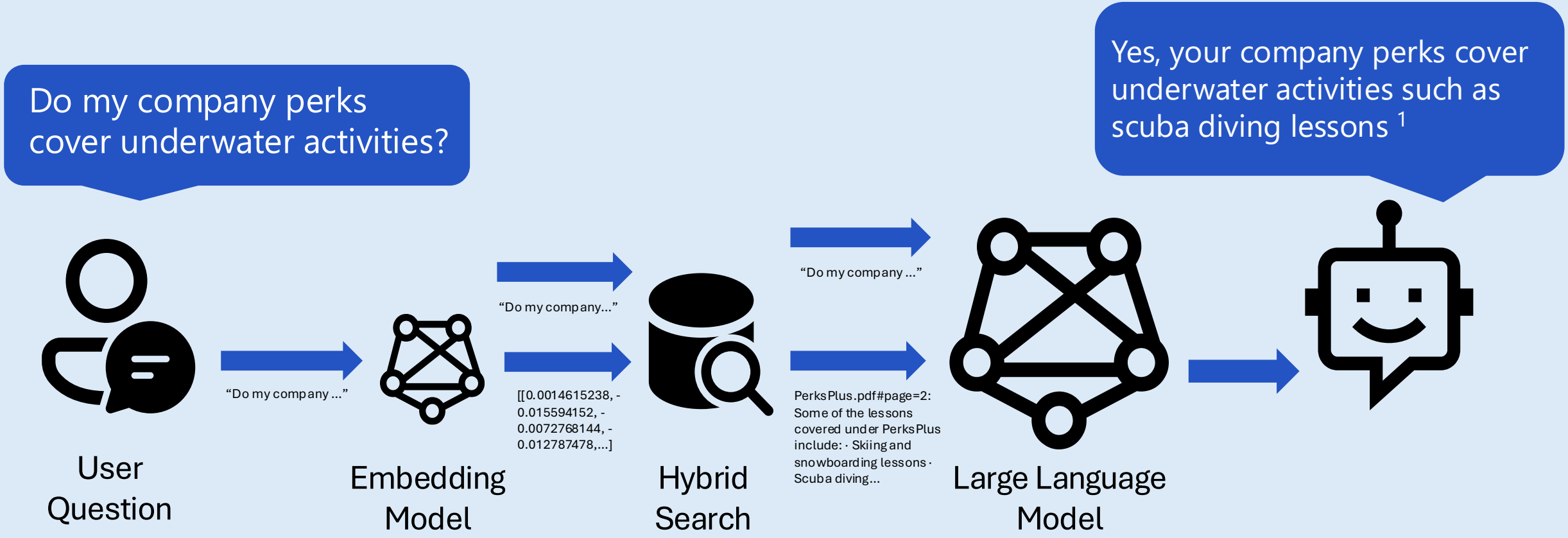
Keyword search is best for exact matches (proper nouns, numbers, etc)

Hybrid search combines vector search and keyword search, optimally using Reciprocal-Rank-Fusion for merging results and a ML model to re-rank results after



https://aka.ms/ragrelevance

RAG with hybrid search



What is the RAG searching?

Documents (Unstructured data)

PDFs, docx, pptx, md, html, images

You need an ingestion process for **extracting**, **splitting**, **vectorizing**, and **storing** document chunks.

On Azure:
Azure AI Search with Document Intelligence, OpenAI embedding models, Integrated Vectorization

Database rows (Structured data)

PostgreSQL, MongoDB, Qdrant, etc.

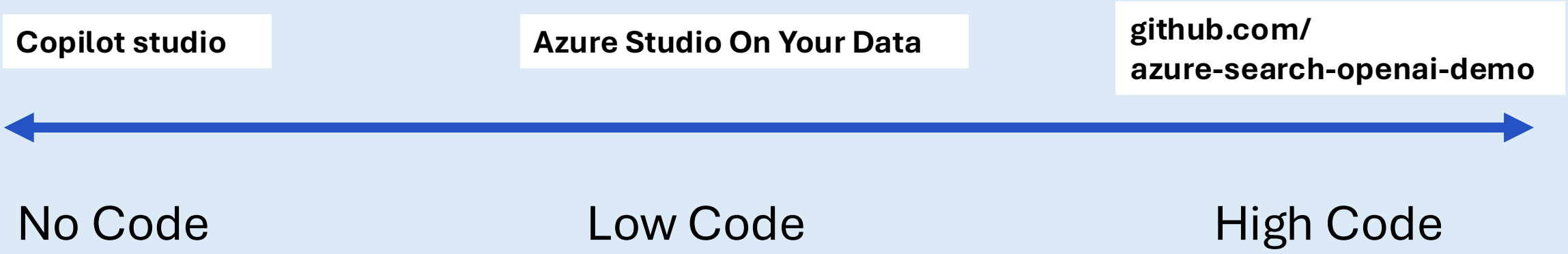
You need a way to **vectorize** & **search** target columns.

On Azure:

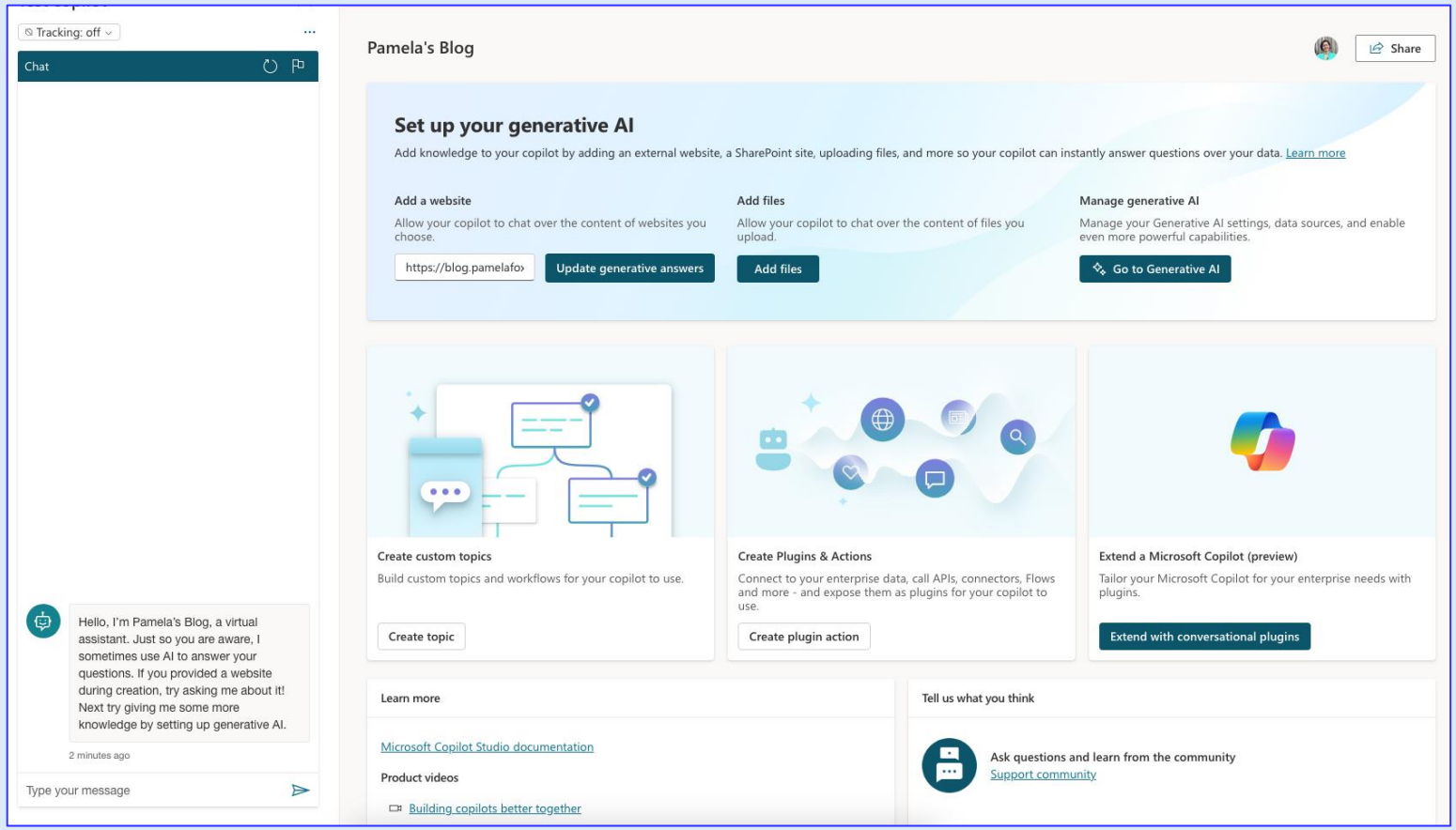
- Azure AI Search (by copying data)
- PostgreSQL+pgvector
- CosmosMongoDB+vector
- Container Apps services (Milvus, Qdrant, Weaviate)

+ OpenAI embedding models

Ways to build a RAG chat app on Azure



Copilot Studio – On Your Data



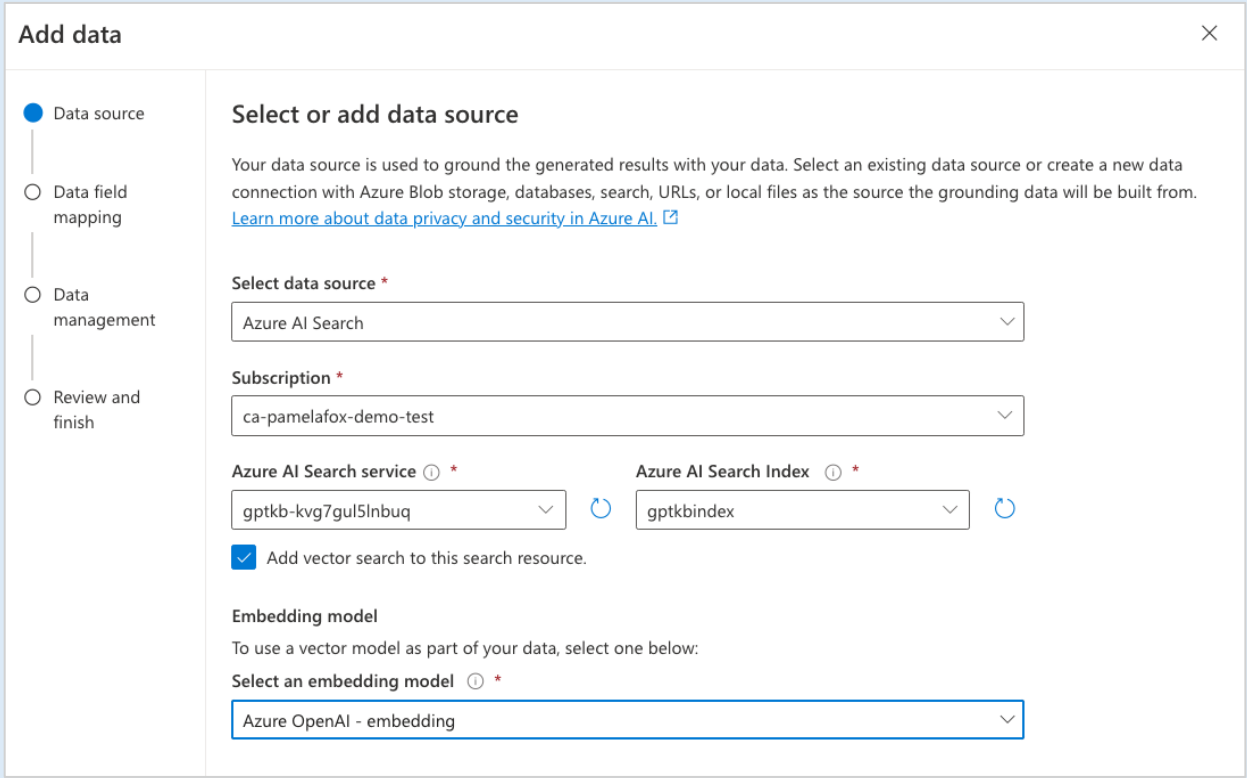
Data type: Documents

Search: ?

LLM: ?

https://copilotstudio.preview.microsoft.com/

Azure Studio – On Your Data



Data type:

Documents (Uploaded, URL, or Blob)
Databases*

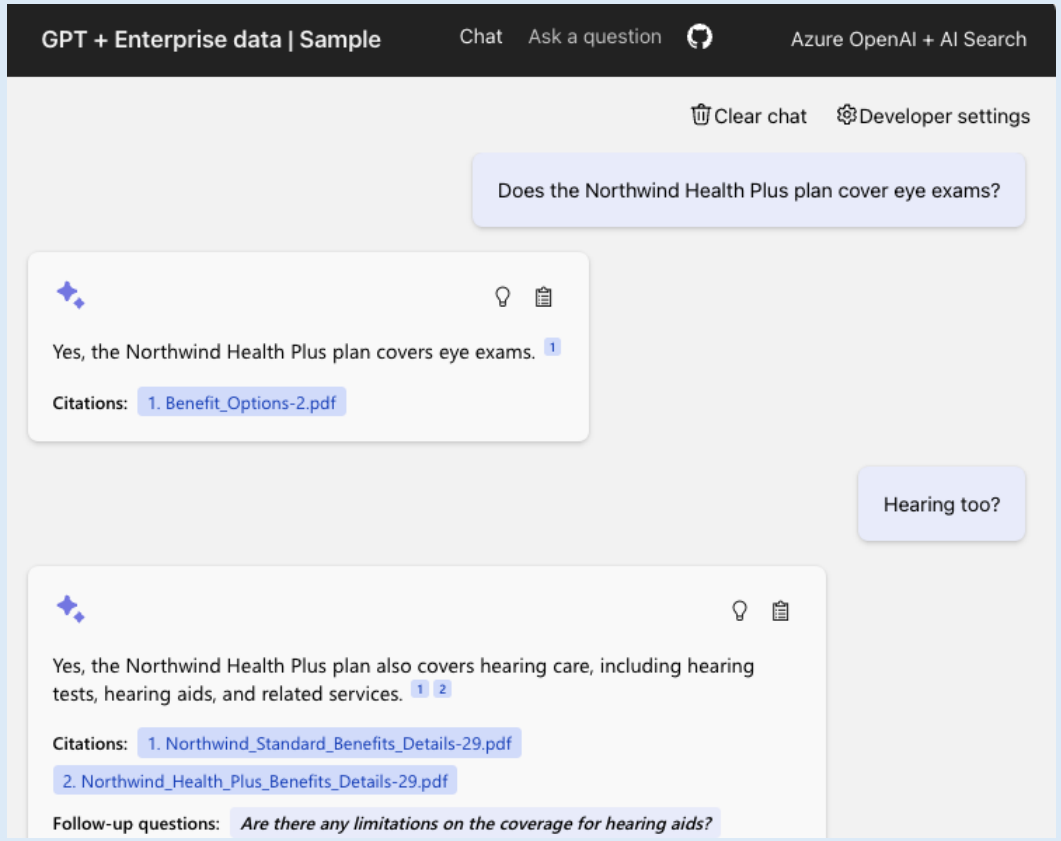
Search:

Azure AI Search
Azure CosmosDB for MongoDB vCore
Azure AI MLIndex
Elastic search
Pinecone

LLM: GPT 3.5/4

https://learn.microsoft.com/azure/ai-services/openai/concepts/use-your-data

Open source RAG chat app solution



Data type:
Documents

Search:
Azure AI Search

LLM: GPT 3.5/4

Features:
Multi-turn chats
User authentication with ACLs
Chat with image documents

https://github.com/Azure-Samples/azure-search-openai-demo/ aka.ms/ragchat

Deep dive: RAG chat app solution

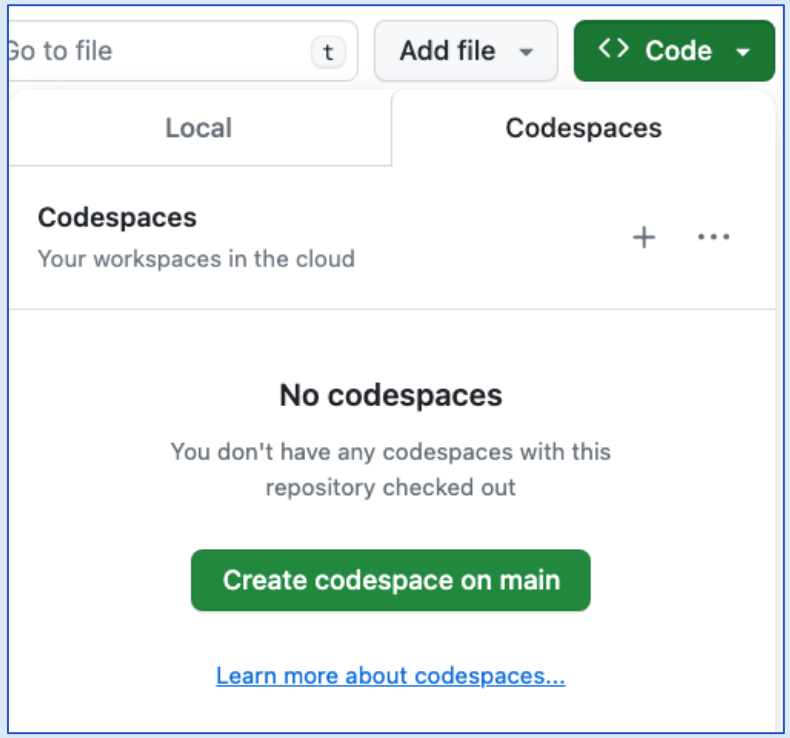
Prerequisites

- Azure account and subscription
  - A free account can be used, but will have limitations.
- Access to Azure OpenAI or an openai.com account
  - Request access to Azure OpenAI today! https://aka.ms/oaiapply
- Azure account permissions:
  - Microsoft.Authorization/roleAssignments/write
  - Microsoft.Resources/deployments/write on subscription level

https://github.com/Azure-Samples/azure-search-openai-demo/#azure-account-requirements

Opening the project: 3 options

- GitHub Codespaces →
- VS Code with Dev Containers extension
- Your Local Environment
  - Python 3.9+
  - Node 14+
  - Azure Developer CLI



https://github.com/Azure-Samples/azure-search-openai-demo/?tab=readme-ov-file#project-setup

Deploying with the Azure Developer CLI

Login to your Azure account:

```
azd auth login
```

Create a new azd environment: (to track deployment parameters)

```
azd env new
```

Provision resources and deploy app:

```
azd up
```

azd up is a combination of azd provision and azd deploy

Application architecture on Azure

CHAT APP

Azure
App Service
or
Local server

Azure
Storage

Azure
OpenAI

Azure
AI Search

DATA INGESTION

Integrated
vectorization
or
Local script

Azure
Document
Intelligence

Azure
Storage

Azure
OpenAI

Azure
AI Search

Local data ingestion

See prepdocs.py for code that ingests documents with these steps:

Azure Blob Storage

Upload documents

An online version of each document is necessary for clickable citations.

→

Azure Document Intelligence

Extract data from documents

Supports PDF, HTML, docx, pptx, xlsx, images, plus can OCR when needed.

Local parsers also available for PDF, HTML, JSON, txt.

→

Python

Split data into chunks

Split text based on sentence boundaries and token lengths.

Langchain splitters could also be used here.

→

Azure OpenAI

Vectorize chunks

Compute embeddings using OpenAI embedding model of your choosing.

→

Azure AI Search

Indexing

- Document index
- Chunk index
- Both

Integrated vectorization

In preview

End-to-end data processing tailored to RAG, built into Azure AI Search

Data source access

- Blob Storage
- ADLSv2
- SQL DB
- CosmosDB
- ...

+ Incremental change tracking

File format cracking

- PDFs
- Office documents
- JSON files
- ...

+ Extract images and text, OCR as needed

Chunking

- Split text into passages
- Propagate document metadata

Vectorization

- Turn chunks into vectors
- OpenAI embeddings or your custom model

Indexing

- Document index
- Chunk index
- Both

Integrated data chunking and embedding in Azure AI Search aka.ms/integrated-vectorization

Code walkthrough

Typescript frontend

(React, FluentUI)

chat.tsx
makeApiRequest()

→

api.ts
chatApi()

Python backend

(Quart, Uvicorn)

app.py
chat()

→

chatreadretrieveread.py
run()

↓

get_search_query()
compute_text_embedding()
search()
get_messages_from_history()
chat.completions.create()

RAG orchestration libraries

Project	Languages
Langchain https://www.langchain.com/	Python, TypeScript, Java
Llamaindex https://docs.llamaindex.ai/	Python, TypeScript
(Microsoft) Semantic Kernel https://github.com/microsoft/semantic-kernel	Python, .NET
(Microsoft) PromptFlow https://github.com/microsoft/promptflow	Python

More RAG chat app starter repositories

GitHub repository	Technologies
Azure-Samples/azure-search-openai-javascript	NodeJS backend, Web components frontend
Azure-Samples/azure-search-openai-demo-csharp	.NET backend, Blazor Web Assembly frontend
Azure-Samples/azure-search-openai-demo-java	Java backend, React frontend
microsoft/sample-app-aoai-chatGPT	Code powering “Azure AI Studio On Your Data”
microsoft/chat-copilot	.NET backend w/Semantic Kernel, React frontend,

Evaluating RAG chat apps

Are the answers high quality?

- Are they correct? (relative to the knowledge base)
- Are they clear and understandable?
- Are they formatted in the desired manner?

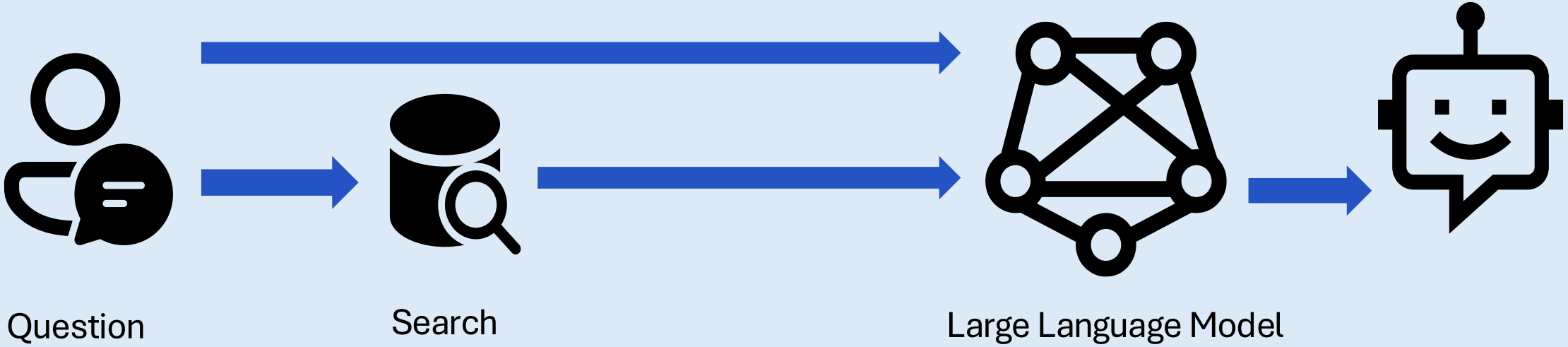
Do the perks cover underwater activities?

Yes, according to the information provided in the PerksPlus.pdf document, underwater activities such as scuba diving are covered under the program.

Yes, the perks provided by the PerksPlus Health and Wellness Reimbursement Program cover a wide range of fitness activities, including underwater activities such as scuba diving. The program aims to support employees' physical health and overall well-being, so it includes various lessons and experiences that promote health and wellness. Scuba diving lessons are specifically mentioned as one of the activities covered under PerksPlus. Therefore, if an employee wishes to pursue scuba diving as a fitness-related activity, they can expense it through the PerksPlus program.

Yes, underwater activities are included as part of the PerksPlus program. Some of the underwater activities covered under PerksPlus include scuba diving lessons [PerksPlus.pdf#page=3].

What affects the quality?

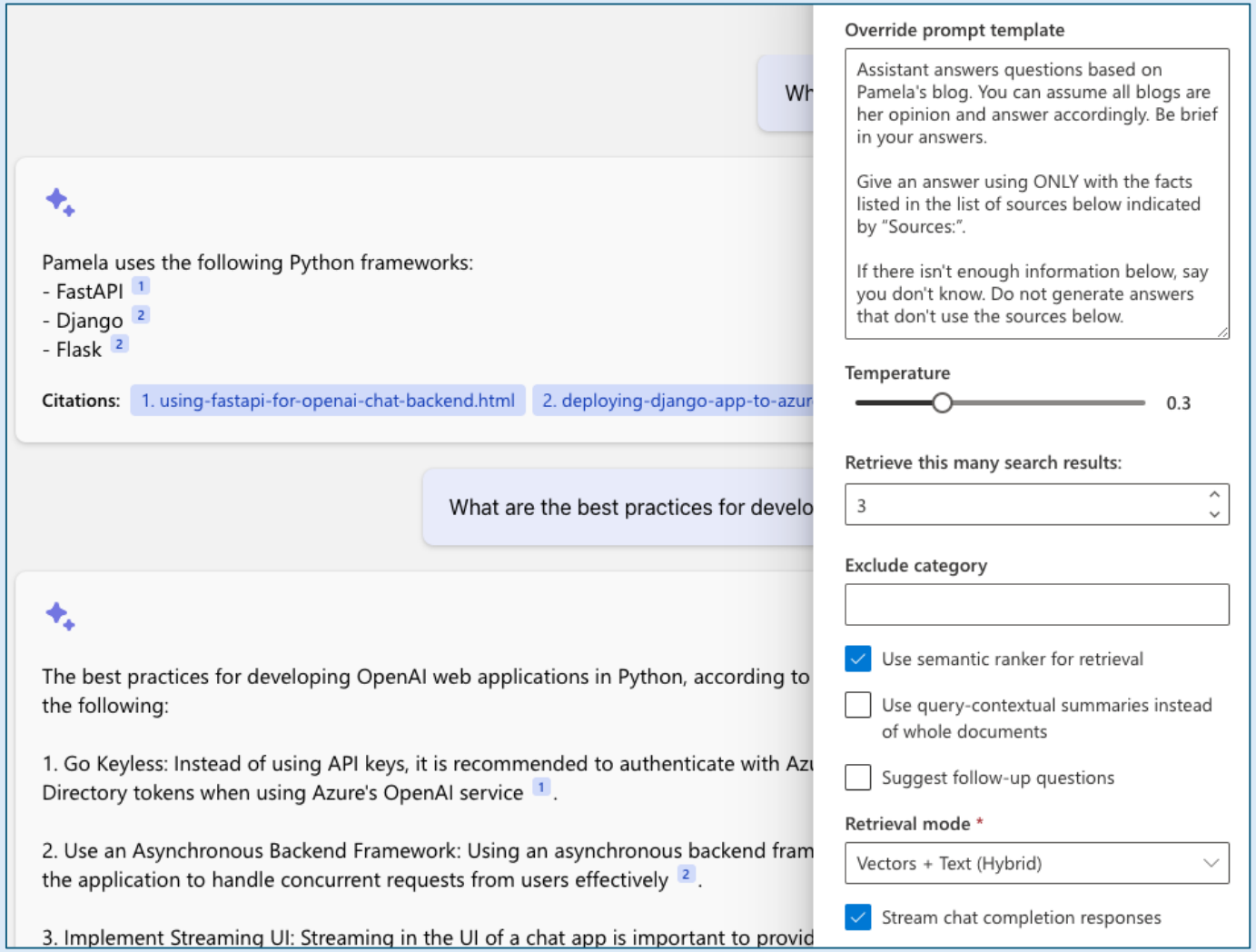


- Search engine (ie. Azure AI Search)
- Search query cleaning
- Search options (hybrid, vector, reranker)
- Additional search options
- Data chunk size and overlap
- Number of results returned

- System prompt
- Language
- Message history
- Model (ie. GPT 3.5)
- Temperature (0-1)
- Max tokens

Manual experimentation of settings

“Developer Settings” →



Automated evaluation of app settings

A set of tools for automating the evaluation of RAG answer quality.

- Generate ground truth data
- Evaluate with different parameters
- Compare the metrics and answers across evaluations

https://github.com/Azure-Samples/ai-rag-chat-evaluator aka.ms/rag/eval

Based on the azure-ai-generative SDK:

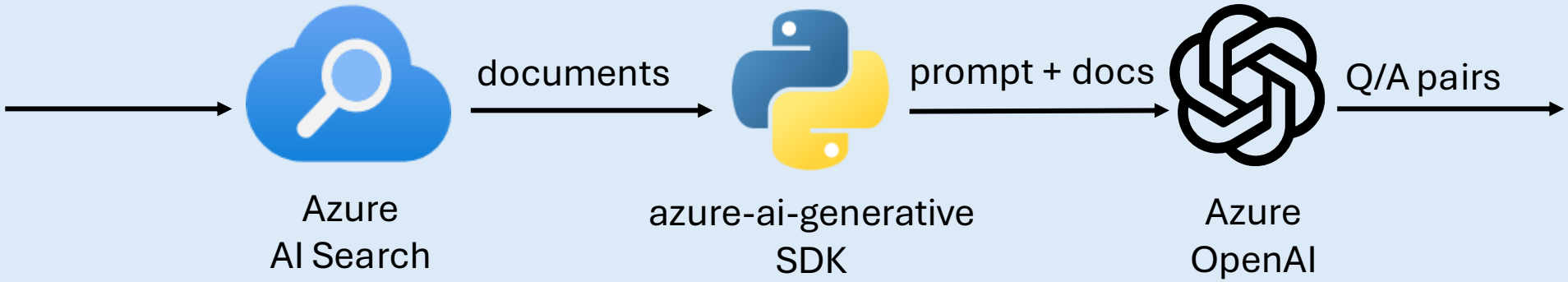
https://pypi.org/project/azure-ai-generative/

Ground truth data

The ground truth data is the ideal answer for a question.
Manual curation is recommended!

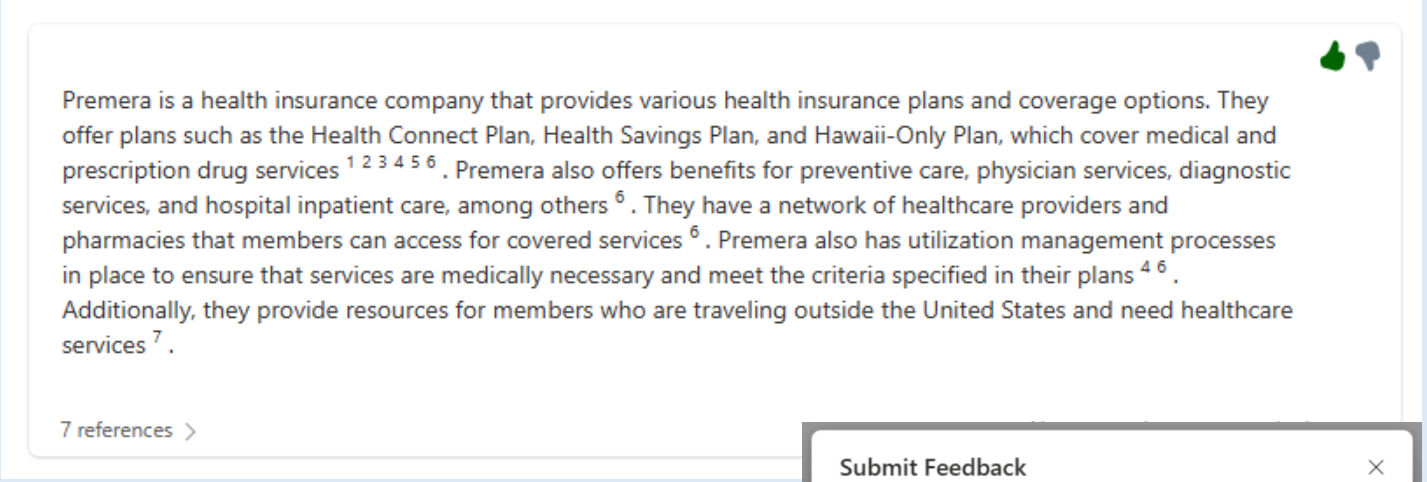
Generate Q/A pairs from a search index:

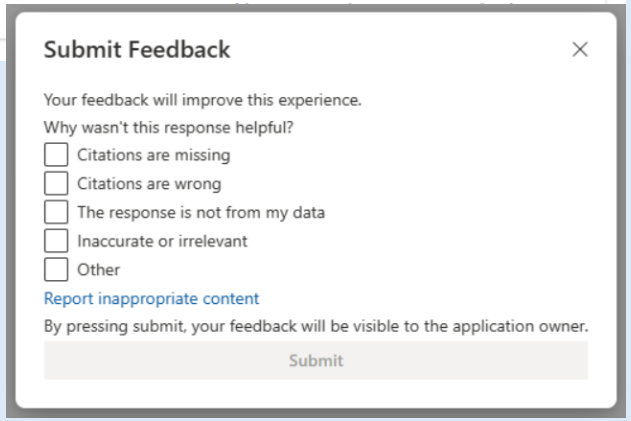
```
python3 -m scripts generate --output=example_input/qa.jsonl
                            --numquestions=200 --persource=5
```



Improving ground truth data sets

Add a 👍/👎 button with feedback dialog to your live app:





Then you can:

- Manually debug the answers that got rated 👎
- Add 👎 questions to ground truth data

https://github.com/microsoft/sample-app-aoai-chatGPT/pull/396

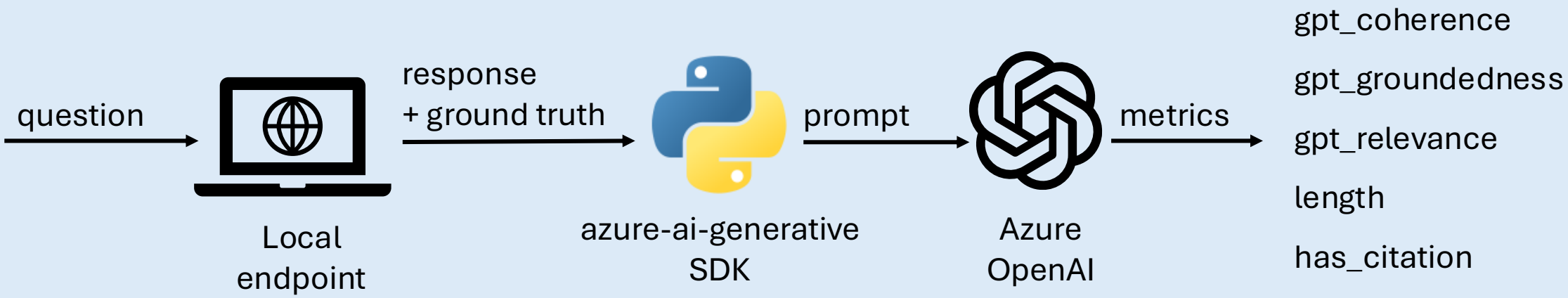
aka.ms/rag/thumbs

Evaluation

Compute GPT metrics and custom metrics for every question in ground truth.

Evaluate based off the configuration:

```
python3 -m scripts evaluate -—config=example_config.json
```



Review the metrics across runs

After you've run some evaluations, review the results:

```
python3 -m review_tools summary example_results
```

folder	groundedness	%	relevance	%	coherence	%	citation %	length
badprompt_ignoresources_1	4.27	0.81	4.10	0.77	4.60	0.89	0.00	1255.44
baseline_1	4.79	0.93	4.78	0.94	4.84	0.96	0.98	648.97
baseline_2	4.89	0.96	4.76	0.94	4.80	0.96	0.98	659.05
baseline_3	4.83	0.94	4.74	0.93	4.80	0.95	0.98	659.39
generalprompt_1	4.86	0.95	4.84	0.96	4.93	0.98	0.00	1164.14
refinedprompt_1	4.93	0.97	4.88	0.96	4.96	0.99	0.99	1430.57
refinedprompt_2	4.87	0.95	4.86	0.96	4.95	0.99	0.99	1398.83
refinedprompt_trimmed_1	4.90	0.97	4.83	0.95	4.93	0.98	0.99	969.89
refinedprompt_trimmed_2	4.91	0.97	4.89	0.97	4.89	0.98	0.99	975.07
refinedprompt_trimmed_3	4.95	0.98	4.84	0.96	4.92	0.98	0.99	965.01
refinedprompt_trimmed_4	4.93	0.97	4.89	0.97	4.86	0.96	0.98	962.78

Compare answers across runs

```
python3 -m review_tools diff example_results/baseline_1
                             example_results/baseline_2
```

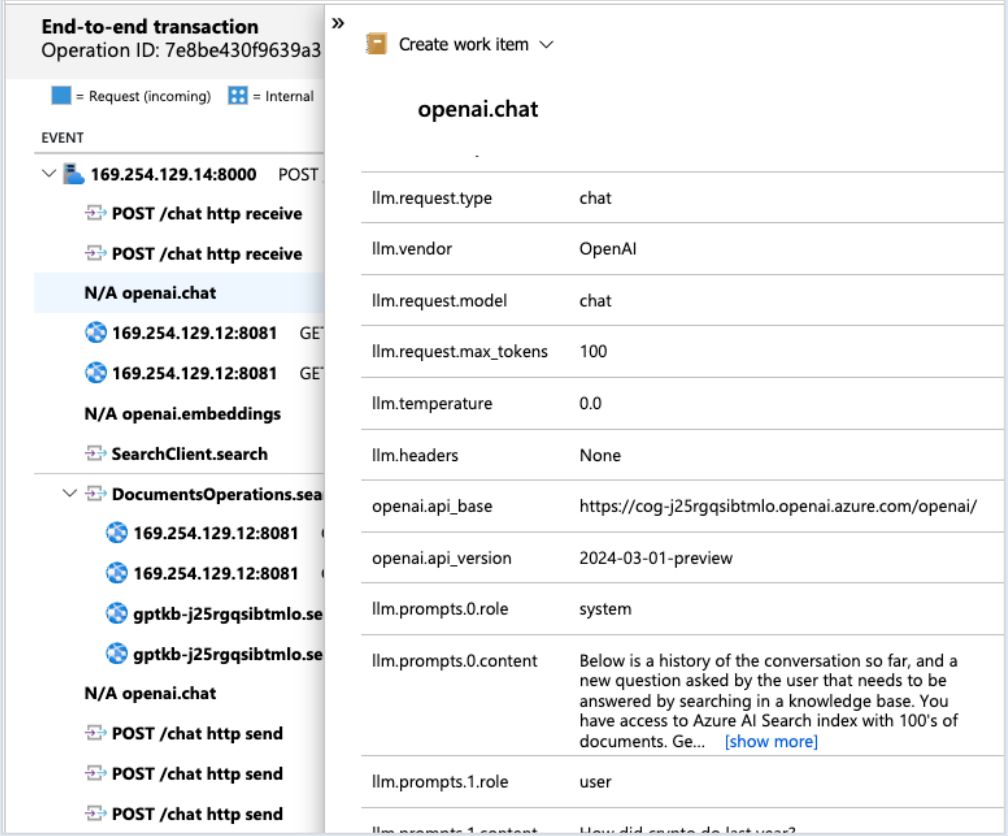
```
What is the importance of choosing the right provider in getting the most value out of your health
insurance plan?

baseline_1                                                  baseline_2

    Choosing the right provider is important                    Choosing the right provider is important
    for getting the most value out of your                      in getting the most value out of your
    health insurance plan. In-network                           health insurance plan for several
    providers charge lower rates than                           reasons:
    out-of-network providers, so choosing an
    in-network provider can help you save                        1. Cost savings: In-network providers
    money on your healthcare costs                                  charge lower rates than
    [Northwind_Standard_Benefits_Details.pdf#                       out-of-network providers, so choosing
    page=5][Northwind_Health_Plus_Benefits_De                       an in-network provider can help you
    tails.pdf#page=4]. In addition,                                 save money on your healthcare costs.
    continuity of care is an important                              [Northwind_Standard_Benefits_Details.

groundedness  relevance  coherence                          groundedness  relevance  coherence
5             5          5                                  5             5          5
```

Observability for RAG chat apps

Integration with Azure Monitor

Send OpenAI traces to Application Insights:

```
if os.getenv("APPLICATIONINSIGHTS_CONNECTION_STRING"):
    configure_azure_monitor()
    # Track OpenAI SDK requests:
    OpenAIInstrumentor().instrument()
    # Track HTTP requests made by aiohttp:
    AioHttpClientInstrumentor().instrument()
    # Track HTTP requests made by httpx:
    HTTPXClientInstrumentor().instrument()
```



https://pypi.org/project/opentelemetry-instrumentation-openai/

Integration with Langfuse

Use the Langfuse wrapper of OpenAI SDK:

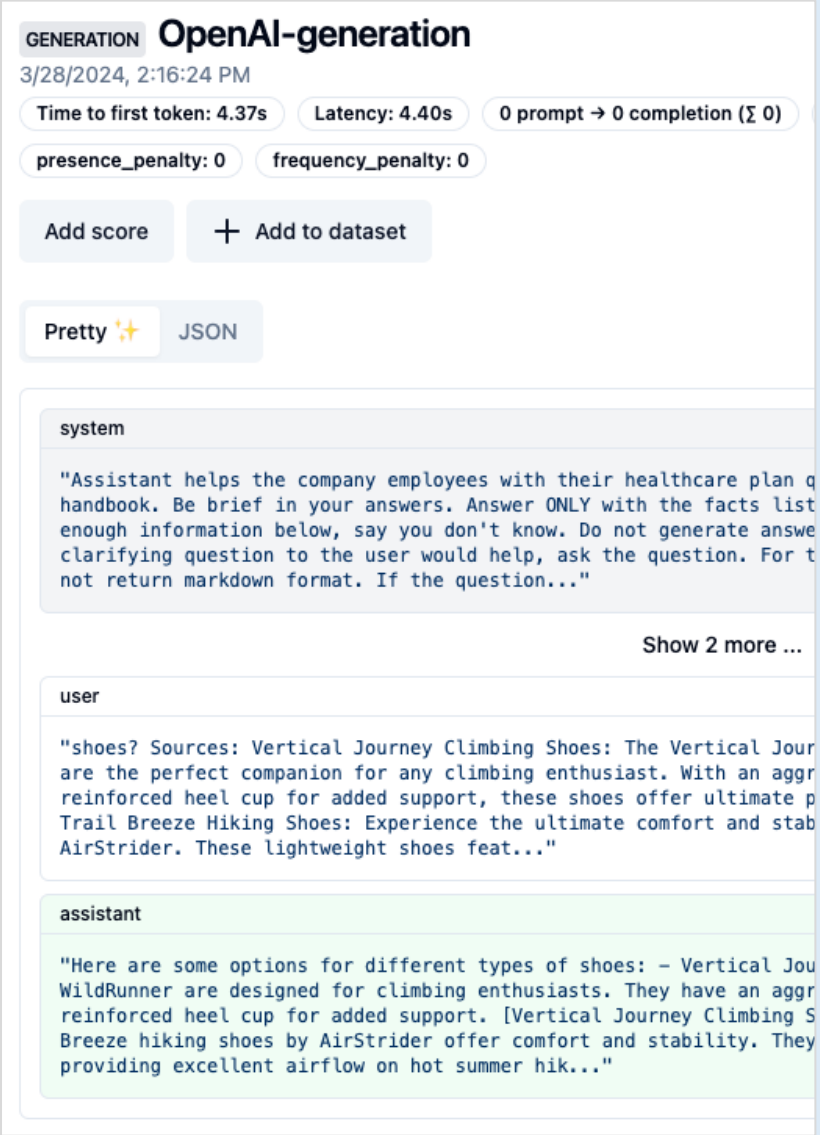
```
if os.getenv("LANGFUSE_HOST"):
    from langfuse.openai import AsyncAzureOpenAI, AsyncOpenAI
```

https://pypi.org/project/langfuse/

Deploy Langfuse to Azure Container Apps + PostgreSQL Flexible:

```
$ azd env set AZURE_USE_AUTHENTICATION true
$ azd up
```

https://github.com/Azure-Samples/langfuse-on-azure/



Next steps

- Create an LLM/RAG app aka.ms/ragchat/free
- Run the evaluator tools aka.ms/rag/eval
- Report any issues or suggest improvements ❔ 🙋🏽‍♂️ 🙋🏾‍♀️
- Share your learnings!